

Applying Machine Learning and the Washburn Equation to Investigate the Biscuit Dunking Process

Artificial Intelligence and Deep Learning
Programming Project 2
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Abstract. This project analyses various datasets exploring the structural properties of McVitie's biscuits and their dynamics during tea dunking. It investigates the effectiveness of machine learning classification and regression algorithms in predicting biscuit type and experimental parameters. The Washburn equation for capillary flow is also evaluated for its predictive accuracy compared to an ML regressor, assessing its applicability to the biscuit-tea interaction. Results show that Support Vector and Multi-Layer Perceptron classifiers effectively identify biscuit type, outperforming Random Forest with accuracies up to 94%. The Washburn equation accurately estimates continuous variables, confirming that tea absorption occurs through capillary flow action. While ML regressors performed reasonably well, they failed to match this Washburn accuracy.

I. INTRODUCTION

A. Biscuit Dunking Experiment and the Washburn Equation

This project analyses the structural properties of three McVitie's biscuits and their dynamics when dunked in tea. Biscuits are composed of interlocking gluten fibres that form a porous structure, facilitating the absorption of liquids via capillary flow action. The Washburn equation is often used in materials science to describe this capillary flow, however, its applicability and accuracy for biscuits remains uncertain. Their complex, irregular porosity and the variable tea viscosity may limit the accuracy of this model:

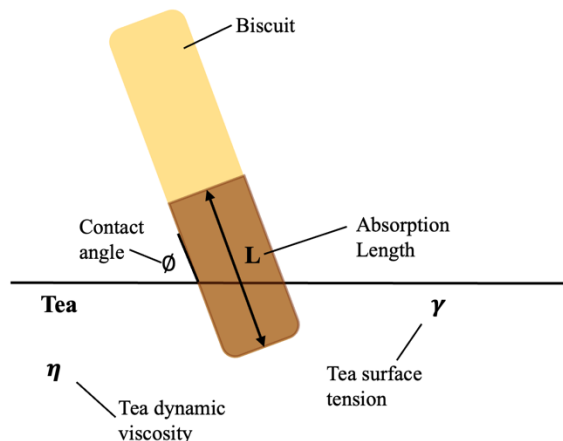
$$L = \sqrt{\frac{\gamma r t \cos(\phi)}{2\eta}} \quad (1)$$

where L represents the distance the liquid (tea) penetrates the solid (biscuit) over a time t , r is the capillary pore radius, ϕ is the contact angle between liquid and solid, while γ and η represent the liquid's surface tension and dynamic viscosity, respectively. Rearranging gives the pore radius as:

$$r = \frac{2\eta L^2}{\gamma t \cos(\phi)} \quad (2)$$

Various datasets were available for analysis: dunking-data, microscopy-data, and time-resolved (TR) data, each containing a set of experimental parameters. The dunking dataset consists of 3000 labelled samples, equally distributed across three biscuit types (Digestive, Hobnob and Rich Tea), including all the Washburn parameters except pore radius. The experimental setup is illustrated in Figure 1. The microscopy dataset is an unlabelled subset of the dunking-data with an additional pore radius feature measured by microscopy. Each of the three TR datasets records the capillary flow rate of a different, unidentified biscuit type, measuring the tea absorption length L over 30-300 seconds. A consistent experimental setup was used for these, controlling surface tension, contact angle and viscosity parameters (Figure 1).

Figure 1. Biscuit dunking experimental setup used to obtain the measurements. Parameters of the Washburn equation are illustrated. Adapted from the project guidelines



B. Aims, Research Questions and Machine Learning Techniques

The project objective is to apply data science techniques to investigate and answer the following research questions:

RQ1: How accurately can a machine learning (ML) algorithm identify different biscuit types, and which parameters are most influential when making these decisions?

RQ2: Is there a significant difference in the pore radius of Digestives, Hobnobs, and Rich Teas?

RQ3: How accurate is the Washburn equation when estimating continuous variables for biscuits, and can ML regression match or exceed this accuracy?

RQ4: Can traditional physics equations and ML algorithms be integrated to enhance model performance?

The analysis done to address these questions is outlined in Tasks 1 to 4 of the accompanying Jupyter Notebook. Several ML algorithms are compared for biscuit classification, including Random Forest, Support Vector Machine, and a Multi-Layer Perceptron (MLP). Random Forest is an ensemble technique that combines the predictions made by multiple decision trees, utilising random feature selection to improve accuracy and reduce overfitting¹ (Figure 2a). Support Vector Classification (SVC) locates the optimal decision boundary (hyperplane) that separates the different classes (e.g. biscuit type) by maximising the distance between the closest points for each class (margin)² (Figure 2b). It is effective in high-dimensional space (lots of features) for both linear and non-linear tasks³. MLP is a simple artificial neural network that incorporates multiple neuron-containing layers to learn complex relationships between input features and output classes.

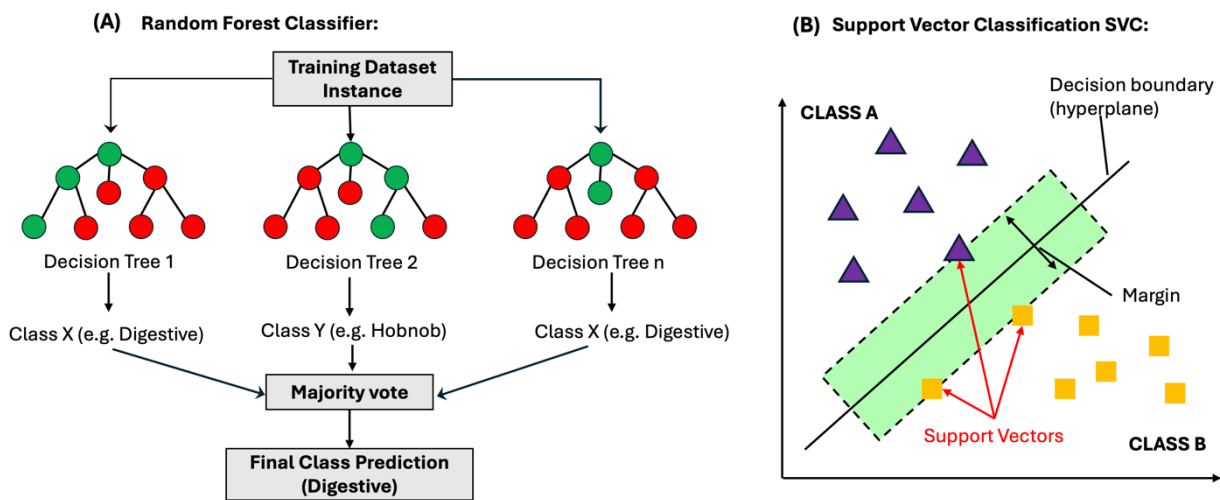


Figure 2. Conceptual diagram showing the simplified workings of (A) Random Forest Classification and (B) Support Vector Classification

II. ANALYSIS AND DISCUSSION

A. RQ1: Comparison of ML Algorithms for Biscuit Classification

To answer RQ1, three classifiers (Random Forest, SVC and MLP) were trained on the dunking dataset and evaluated using various metrics to determine the best performer. First the dunking-data was pre-processed, meaning it was checked for missing values, scaled, and split into training (80%) and testing (20%) datasets. Stratified sampling was used to maintain an equal class distribution (i.e. equal portion of biscuit types) in all datasets. The models were trained using Scikit-Learn’s default hyperparameters, achieving test accuracies of 81.3%, 86.8%, and 93.8% for Random Forest, SVC and MLP, respectively. Optimising each model by tuning their hyperparameters enhanced classification performance. The SVC achieved a significant increase in test accuracy to 93.0%, while the Random Forest and MLP resulted in only minor changes (Figure 3). The improvement in SVC performance was likely driven by the higher regularisation parameter, which increases model complexity with a tighter decision boundary.

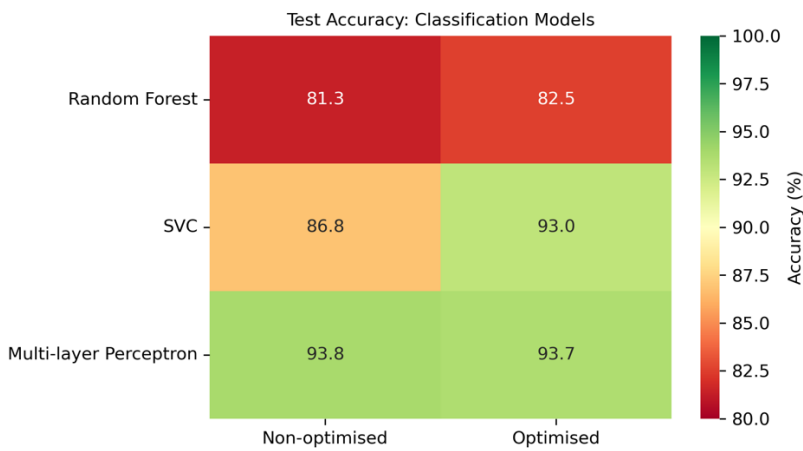


Figure 3. Heatmap showing the improvement in test accuracy after undergoing hyperparameter tuning for each of the classification models. The Support Vector Classifier (SVC) exhibits significant improvement, however, the MLP classifier still has the best performance when predicting biscuit type.

Table 1. Full set of evaluation metrics for the optimised classification models

ML Classifier	Biscuit Type	Precision	Recall	F1-score	Test Accuracy (%)
Random Forest	Digestive	0.86	0.85	0.85	82.5
	Hobnob	0.74	0.73	0.74	
	Rich Tea	0.87	0.90	0.88	
SVC	Digestive	0.96	0.98	0.97	93.0
	Hobnob	0.92	0.89	0.90	
	Rich Tea	0.92	0.93	0.92	
MLP	Digestive	0.97	0.98	0.98	93.7
	Hobnob	0.93	0.88	0.90	
	Rich Tea	0.91	0.95	0.93	

Overall, the MLP and optimised SVC are very effective at predicting biscuit types from these data, outperforming the Random Forest in all metrics (precision, recall, F1-score) (Table 1). This may be due to their superior ability to capture complex patterns or the Random Forest’s tendency to capture/overfit on noise and irrelevant patterns. Interestingly, all models found Hobnobs the hardest to classify (Table 1), indicating it may have greater feature variability and/or overlap with other biscuit types. To assess which features were most important for the model’s predictions, permutation importance was computed for each model (Figure 4).

Permutation importance was chosen over impurity-based methods as it is model-agnostic, enabling valid comparison between classifiers. It measures the performance reduction when each feature is randomly shuffled, with a larger loss indicating greater importance. All classifiers exhibit the same feature importance ranking, showing the absorption length L and contact angle ϕ as most important, and tea viscosity η as the least. An attempt was made to simplify the model and enhance SVC performance by removing the viscosity feature, but it made little difference, reducing accuracy to 92.3%.

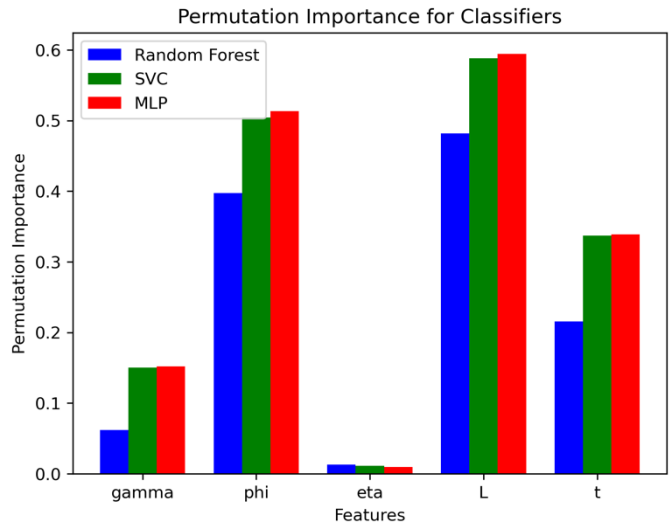


Figure 4. Permutation importance for the three classification models. This shows how important each feature is when making predictions for the type of biscuit.

B. RQ2: Analysing the Difference in Pore Radius across Biscuit Types

The microscopy-data, a subset containing pore radius measurements, was merged with the dunking dataset, aligning the shared parameters. This provided a simple method for assigning biscuit type labels to each sample. Pore radius statistics for each biscuit could then be calculated and are presented in Table 2. Digestives have the largest mean pore radius (0.803 μm), followed by Hobnobs (0.496 μm), then Rich Teas (0.304 μm). Hobnobs exhibit the greatest variability in pore radius, evidenced by its higher standard deviation, while Rich Teas show the least. Figure 5 visualises the distribution, highlighting the significant difference between the three biscuit types.

Table 2. Measured pore radius statistics for each biscuit type (microscopy-data).

Biscuit Type	Count	Mean (μm)	Standard Deviation (μm)	Minimum (μm)	Maximum (μm)
Digestive	168	0.803	0.058	0.700	0.899
Hobnob	163	0.496	0.080	0.232	0.733
Rich Tea	169	0.304	0.047	0.156	0.429

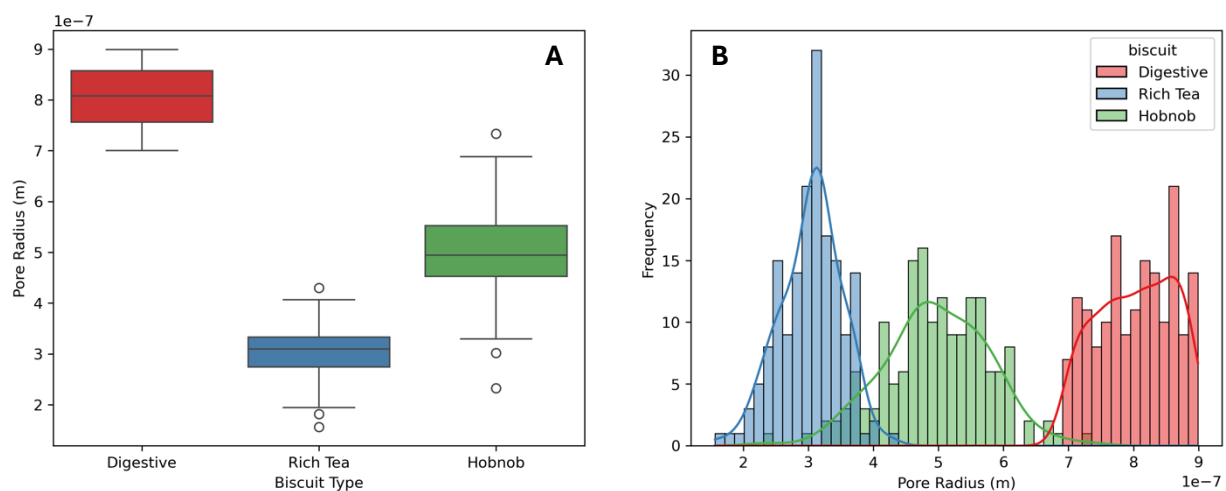


Figure 5. Boxplot and histogram visualizing the difference in pore radii between the biscuit types. These are the measured values from the microscopy data set.

C. RQ3: Evaluating the Performance of the Washburn Equation vs. ML Regression

Two ML regression algorithms, Random Forest and Support Vector Regressor (SVR), were trained on the pre-processed microscopy-data and evaluated on how well they predicted absorption length. Biscuit type was not included in training to ensure a fair comparison to the Washburn equation. The Random Forest significantly outperforms the SVR, yielding a smaller root mean squared error (RMSE) and an impressive R^2 score (0.97) on the test data. RMSE measures the average error between predicted and actual values, while R^2 indicates how well the model explains the variance in the target variable. Together, they provide a comprehensive assessment of model performance. Using the same test data, the Washburn equation was used to predict absorption length and pore radius. Table 3 and Figure 6 compare the Washburn and Random Forest performances for both variables, demonstrating the Washburn equation's superior accuracy with lower RMSE and higher R^2 scores all round. While the Random Forest predicts absorption length well, it struggles with predicting pore radius ($R^2=0.71$), shown by the deviation from the 'perfect prediction' line in Figure 6b. Figure 7 further highlights this, showing the Random Forest failing to capture the three regions representing biscuit types. The Washburn equation, however, successfully predicts a distribution similar to the actual values, with clear separation of biscuits. In summary, the Washburn equation is very accurate in predicting both variables, proving its applicability to biscuits. Although the ML regressor performed reasonably well, it was not able to match this accuracy.

Table 3. Comparison of how well the Random Forest regressor and the Washburn Equation predict continuous variables L and r.

Variable	Method	RMSE (m)	R^2
Absorption Length (L)	Random Forest regressor	4.98×10^{-4}	0.9686
	Washburn equation	6.84×10^{-5}	0.9994
Pore Radius (r)	Random Forest regressor	1.15×10^{-7}	0.7124
	Washburn equation	1.19×10^{-8}	0.9969

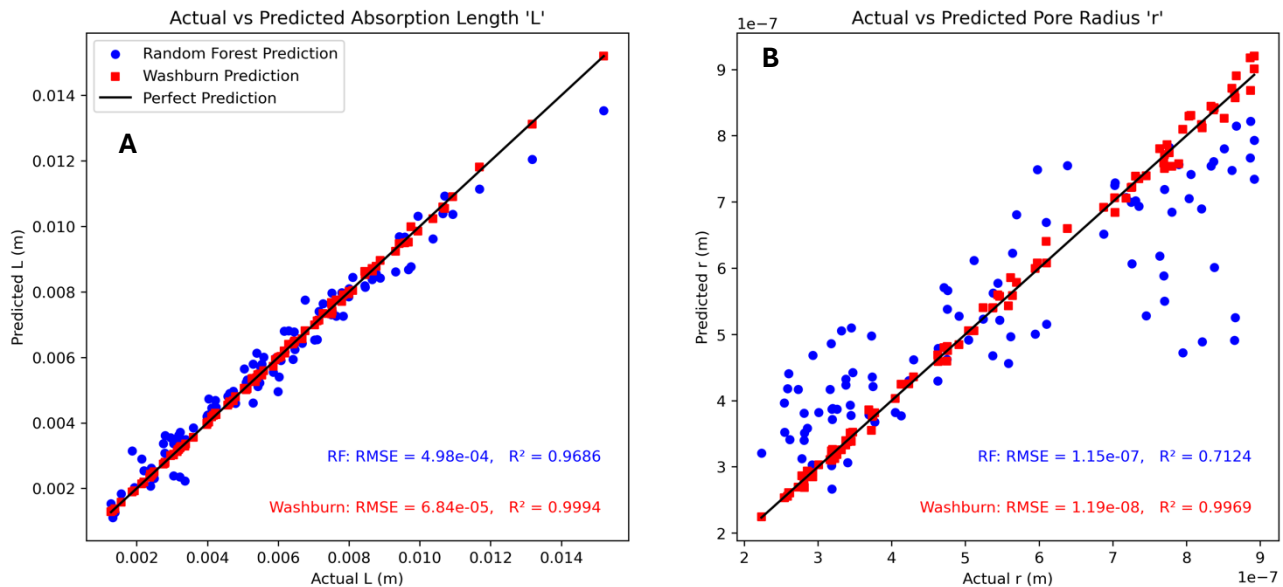


Figure 6. Scatterplots comparing the performance of the Washburn equation and Random Forest regressor when predicting continuous variables. **(A)** Shows that both methods predict absorption length 'L' very well, evidenced by their tight fit to the 'perfect prediction' line (black). **(B)** Highlights that the Random Forest struggles to predict pore radius 'r' and is significantly outperformed by the Washburn equation. The Washburn equation is highly accurate for estimating both variables.

Distributions of the predicted pore radii (r) vs the actual values

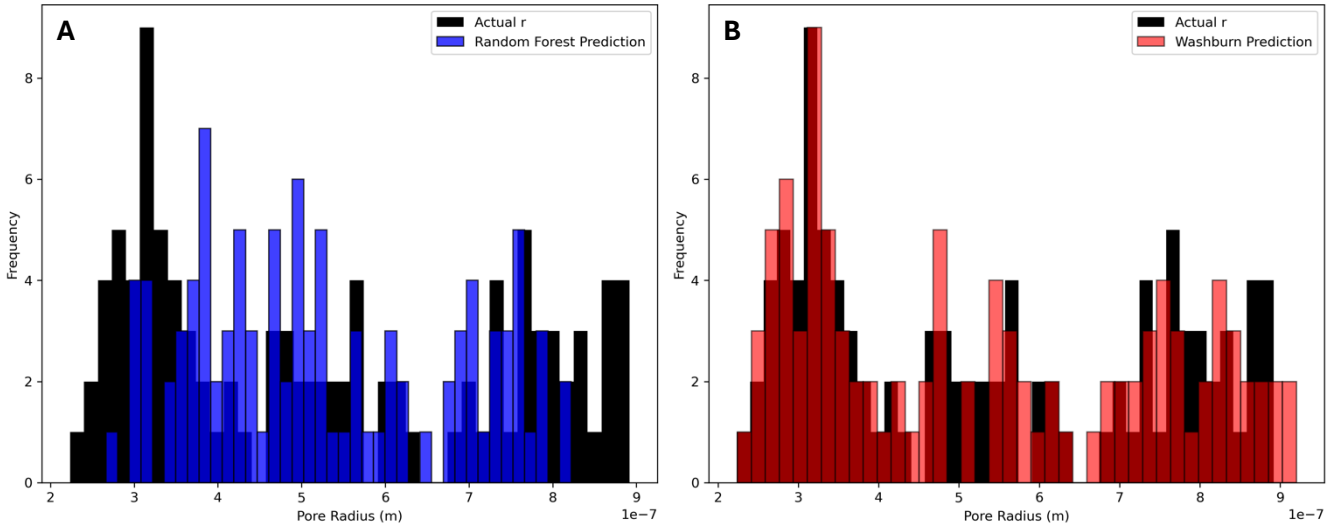


Figure 7. Histograms showing the predicted pore radius distribution vs the actual pore radius distribution (black). (A) Random Forest prediction (blue) does not match the distribution of the actual values and fails to capture the three spikes representing the biscuit types. (B) In contrast, the Washburn equation (red) successfully predicts a similar distribution, capturing the three portions much better.

D. RQ4: Enhancing the Classification Model by Incorporating a Physics-derived Feature

An attempt was made to identify the biscuit type used for each TR dataset using the SVC model trained on the dunking-data from Task 1. This was unsuccessful, as shown by the cross-over of biscuit types and datasets (Figure 8). This is likely due to the different time ranges between the datasets and/or the fixed contact angle, given its high permutation importance.

Given its high accuracy in predicting pore radius r for RQ3, the Washburn equation was used to estimate r for the dunking dataset. The SVC was then retrained with r as an additional feature, increasing biscuit classification accuracy from 93.0% to 95.5%. Recalculating permutation importance suggests that this new model relies almost entirely on r for predicting biscuit type, making the other features redundant (Figure 9). This indicates a dominant correlation between pore size and biscuit type. We then calculate r for the TR-data and predict biscuit type using the new model. This successfully identifies the biscuit for each dataset (Figure 10).

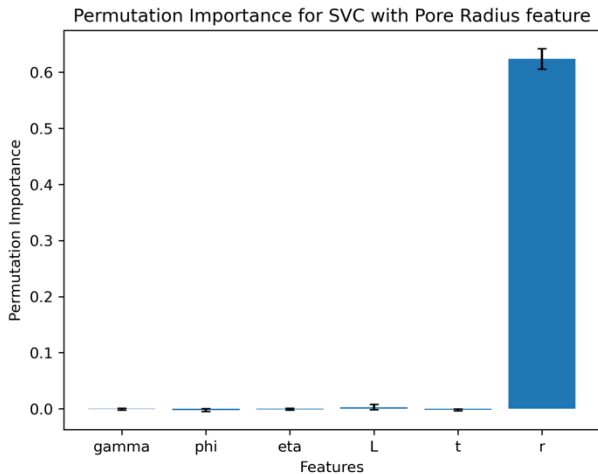


Figure 9. Permutation importance for the new SVC model trained with an additional pore radius input feature.

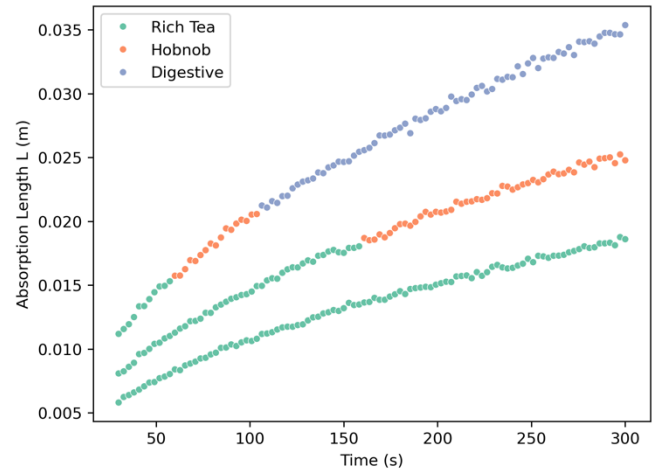


Figure 8. Absorption length (L) vs time for all three time-resolved datasets (tr-1, tr-2, tr-3). Each line is a different biscuit type.

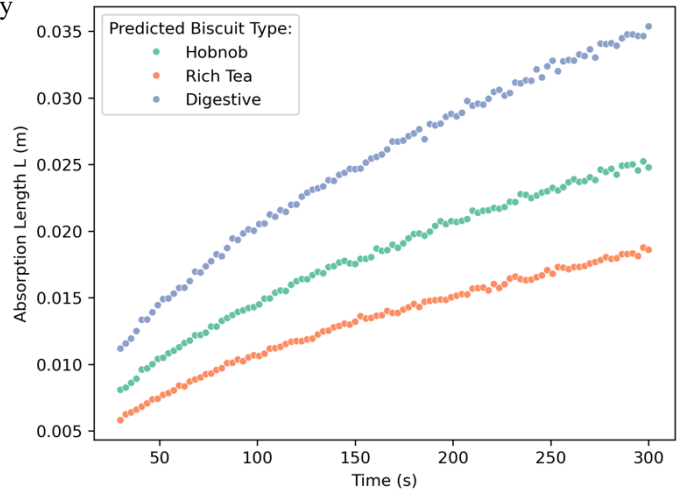


Figure 10. Absorption length (L) vs time for all three time-resolved datasets. Predictions for biscuit type made by the new SVC model, trained with an additional pore radius feature, are now successful. tr-1 = Hobnob, tr-2 = Rich Tea, tr-3 = Digestive.

E. Additional Analysis: Which Variable has the Largest Impact on Absorption Length?

Two approaches were used to answer this question: 1) Training a Random Forest regressor on the dunking-data, predict absorption length, and visualising the feature (Gini) importance scores. 2) Calculating the Pearson Correlation Coefficients to assess the linear relationship between each variable and L . Both methods found consistent results, indicating that contact angle ϕ has the strongest relationship with L , followed by pore radius, then time (Figure 11). Having the highest feature importance indicates ϕ contributes most to L predictions, which can be interpreted as having the largest impact on L . This is supported by its correlation coefficient (-0.79), revealing the two variables have a strong negative correlation (i.e. as contact angle decreases, absorption length increases). Tea surface tension and viscosity have the least influence.

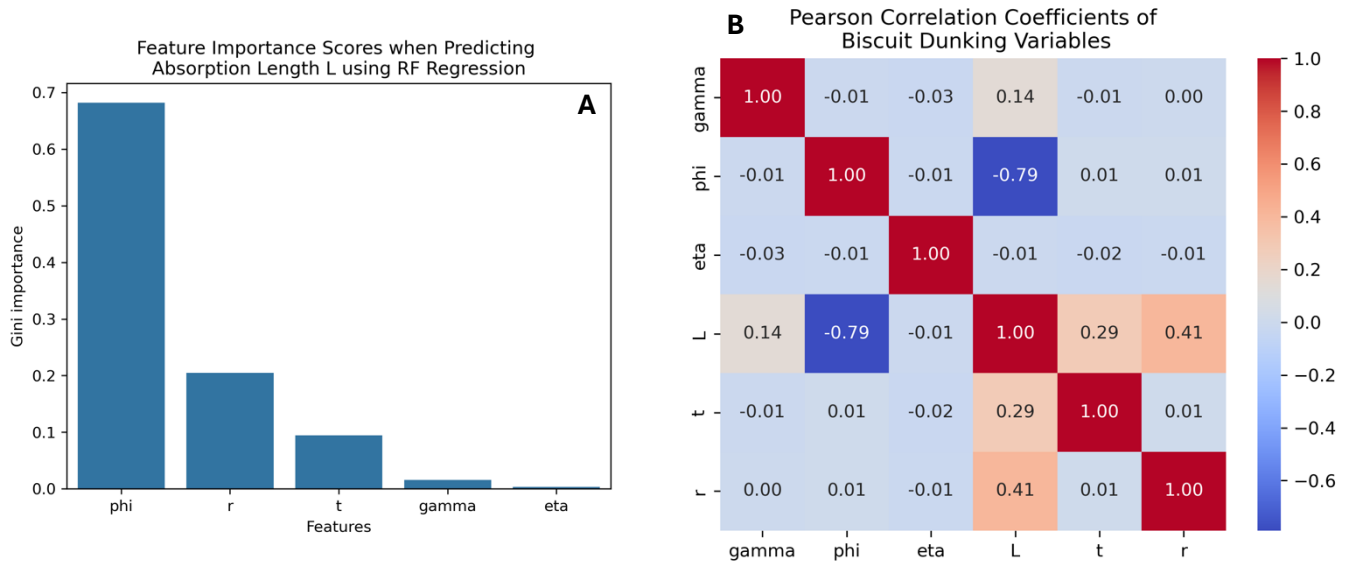


Figure 11. (A) Feature Importances when predicting L using a Random Forest regressor trained on the dunking-data. (B) Heatmap of the Pearson correlation coefficient assessing the relationships between each variable and L .

III. CONCLUSION

This analysis demonstrates that ML classification algorithms can effectively identify biscuits, achieving accuracies up to 94%. However, ML regressors struggle to predict continuous variables, such as L and r , with the same accuracy as the Washburn equation. The low RMSE and high R^2 scores obtained by the Washburn equation prove its applicability to biscuits and confirm that tea absorption occurs via capillary flow action. Other results show a significant difference in pore size between biscuit type, and that the biscuit-tea contact angle has the largest impact on absorption length. Several improvements/additional analyses could be done to enhance this investigation. For example, the uncertainties in the TR-data could be used in probabilistic models/techniques, such as weighted loss functions and Bayesian regression models. These could enhance model robustness and overall performance. Incorporating more variables such as biscuit density and tea temperature would also offer deeper insights into the biscuit-dunking process.

REFERENCES

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